中山大學
SUN YAT-SEN UNIVERSITY

Yongxue Xu (Jerry)

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Summary

Undergraduate researcher at Sun Yat-sen University working on video understanding and generation, multimodal foundation models, world models, and 4D scene understanding. I am particularly interested in models that connect perception, generation, and interaction, and I am currently seeking industry research internships in these areas.

Education

Sun Yat-sen University (SYSU) *Project 985 / 211 / Double First-Class* Sep. 2023 – Jun. 2027
B.Eng. in Intelligent Science and Technology, School of Intelligent Systems Engineering
GPA: 3.91 / 4.0 | **Major GPA:** 3.94 / 4.0 *(Major Required + Major Elective)*

Skills

Research	Video understanding and generation; multimodal foundation models; world models; 4D scene understanding; world-action models.
Programming	Python, C/C++, MATLAB; experienced with PyTorch.
Languages	Chinese (native); English (CET-6, academic reading and writing).

Research Experience

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

- Overview: Built a multi-view temporal benchmark spanning three domains and proposed **LMM-Track4D** for object-level 4D reasoning. Trajectory-grounded dialogue, a streaming [TRK] state token, and trajectory decoding support both textual QA and 3D trajectory prediction.
- Responsibilities: Led the technical roadmap, model design, and main experiments; contributed to benchmark construction.
- Outcome: Submitted to **NeurIPS 2026** as **co-first author**.

Learning to Reason with Persistent Object States for Video Instance Segmentation

- Overview: Proposed **POSReasoner**, a trainable plug-and-play framework that reasons over persistent object states. A sparse state–observation graph and Propose–Verify procedure resolve competing associations and selectively retain, update, or reactivate states while the host model remains frozen.
- Responsibilities: Led the technical roadmap, architecture, implementation, experiments, and manuscript development.
- Outcome: Preprint; **co-first author**.

Graph Domain Adaptation Does Not End with Representation Learning

- Overview: Proposed **EviGDA**, pairing an entropy-aligned graph-aware expert with an independent graph-free expert. Post-training fusion exploits complementary evidence without learned routing or target pseudo-labels.
- Outcome: State-of-the-art results on ten datasets and 16 transfer tasks; preprint; **co-first author**.

Robust Competitive Influence Maximization on Multilayer Networks

- Overview: Proposed a diffusion-aware, role-guided evolutionary framework combining cross-layer alignment, role constraints, dynamic robustness evaluation, and multi-task evolutionary search.
- Responsibilities: Defined the roadmap; designed core modules and experiments; led patent filing and manuscript development.
- Outcome: **1 patent pending**; manuscript under review at **IEEE TNSE** as **first author**.

Critical Node Identification in Urban Road Networks via Deep Reinforcement Learning

- Overview: Evaluated deep and reinforcement learning methods on real urban road networks, using GNNs for topology representation and DQN-based agents for adaptive critical-node decisions. Contributed dataset construction, baseline reproduction, and visualization.
- Outcome: **Third author**; paper received the **Best Paper Award** at **GBCESC 2025**.

Selected Publications

MANUSCRIPTS AND PREPRINTS

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue
Chaoyue Li*, **Yongxue Xu***, Jie Feng*, and Jiayu Ding†
Submitted to NeurIPS 2026 2026

Learning to Reason with Persistent Object States for Video Instance Segmentation
Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang†
Preprint 2026

Graph Domain Adaptation Does Not End with Representation Learning
Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang†
Preprint 2026

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach
Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang†
Under review at IEEE Transactions on Network Science and Engineering 2026

CONFERENCE PAPER

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks
Zhonghan Liu, Wenshuo Liu, **Yongxue Xu**, Ziqian Liu, Yichen Liu, and Shuai Wang†
GBCESC 2025, Best Paper Award 2025

* Equal contribution. † Corresponding author.

Research Initiatives & Collaborations

- Collaborate with researchers at **HKUST**, the **EPIC Lab at SJTU SAI**, and **Westlake University** on video generation, world-action modeling, and multimodal spatiotemporal understanding.
- Co-organize **InkMind.AI**, a small research group exploring multimodal understanding and generation. We also study vibe-coding workflows for recursive self-improvement and automated research, with the goal of making AI-assisted research more precise, efficient, and insight-driven.

Awards & Competitions

Chinese Mathematics Competition (CMC), Guangdong Region — Second Prize Dec. 2024
Mathematical Contest in Modeling (MCM), USA — Honorable Mention Feb. 2025
National Undergraduate Smart Car Competition, South China Region — Second Prize Aug. 2025

Interests

Table tennis, music, and nature travel.